INCH-POUND

MIL-S-81746/2
5 April 1991

MILITARY SPECIFICATION SHEET

SERVTORQ, TYPE 23ST64A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Servtorqs described herein shall consist of this specification sheet and the issue of the following specification listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-81746.

TABLE I. Requirements.

Requirement	Value	Unit	Tolerance
Reference voltage 1/	115	volts	±1%
Frequency	50-400	hz	±1%
Power:			
Primary	5.0	watts	maximum
Stall	20	watts	maximum
Current at 60 Hz:			
Primary	50	milliamps	maximum
Stall	150	milliamps	maximum
Electrical error	10	minutes	maximum
Stall torque:			
At 60 Hz	4.4	ounce-inches	minimum
At 400 Hz	5.3	ounce-inches	minimum
No-load slewing speed	500	rpm	minimum
Servtorq sensitivity	6.0	minutes	maximum
Dynamic error	30	minutes	maximum
Synchronizing time:			
30° ±2°	0.5	seconds	maximum
177° ±2°	1.0	seconds	maximum
183° ±2°	1.0	seconds	maximum
330° ±2°	0.5	seconds	maximum
Slewing speed under load	450	rpm	minimum
Temperature rise:			
Quiescent	16	degrees (C)	maximum
Stall	24	degrees (C)	maximum
Torque gradient:		oz-in per degree	minimum
At 60 Hz	20	oz-in per degree	minimum
At 400 Hz	24	oz-in per degree	minimum
Shaft load inertia	0-3000	gm-cm^2	min.-max.

FSC 5990

AMSC - N/A

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TABLE I. Requirements - Continued.

Requirement	Value	Unit	Tolerance
Velocity constant:			
At 60 Hz	13,000	$seconds^{-1}$	minimum
At 400 Hz	17,000	$seconds^{-1}$	minimum
Variation of voltage (±10%) & frequency (±5%)	6.0	watts	maximum
Radial play	0.0010	inches	maximum
End play	0.0020	inches	±0.0010
Weight	20	ounces	maximum

NOTES:

1/Type 23ST64A servtorqs shall be capable of operating in ambient temperatures ranging from -55°C to +85°C.

TABLE II. Part identifying number variant characteristics.

Part Identifying Number 1/	A ±.020	B ±.020	G Maximum	Shaft Style	Terminal End	U +.0000 -.0003
M81746/2-01A	.676	.719	3.576	Threaded	Terminal Board	.2405

1/Part identifying number M81746/2-01A shall be the NATO standard.

1/The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

Custodian:
Navy-AS

Preparing Activity:
Navy-AS

Review Activities:

(Project No. 5990-0396-02')

User Activities:
Navy-SH,EC

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